

10 PEARLS AQI PREDICTOR

An End-to-End Serverless MLOps Pipeline for 3 Days Air Quality Forecasting

A serverless, end-to-end MLOps pipeline that predicts the Air Quality Index three days ahead for six major Pakistani cities fully automated, from data collection to a live dashboard.



The Challenge, and How I Solved It

Real problems encountered along the way, and the engineering decisions that addressed them

THE PROBLEM

1 No live data for Cities

AQICN's own station-search and map APIs confirmed there was no active ground station covering Karachi or Sindh at all.

2 Feature store was down

Hopsworks, the original plan, reported an active platform outage affecting data ingestion during setup.

3 No automation or trust

Manually re-running scripts meant no schedule, no failure handling, and results going stale within hours.

THE SOLUTION

1 Switched to OpenWeather API

OpenWeather's satellite-based Air Pollution API gave reliable, current coverage for all six target cities, plus real historical data.

2 Used Vertex AI instead

BigQuery as the feature store and Vertex AI Model Registry for trained models the other option the brief allowed.

3 GitHub Actions, fully automated

Hourly feature collection and daily retraining, authenticated keylessly via Workload Identity Federation.

Tech Stack: Data Collection & Modeling

Where the numbers come from, and how the forecast gets made



DATA SOURCE

OpenWeather API

Live weather and pollutant concentrations for six cities, plus 30 days of real hourly historical data for training.



LANGUAGE

Python

Every pipeline, training script, and application in the project is written in Python.



STATISTICAL MODELS

Scikit-learn

Ridge Regression and Random Forest, trained per city and per forecast day, evaluated with RMSE, MAE, and R^2 .



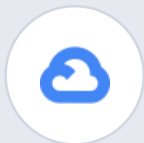
DEEP LEARNING

TensorFlow

A custom neural network competes against the statistical models; predictions are combined by median for robustness.

Tech Stack: Cloud, Serving & Automation

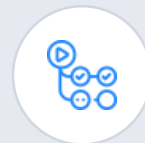
Where the data and models live, and what keeps everything running



FEATURE STORE & REGISTRY

Vertex AI

BigQuery holds the feature store; Vertex AI Model Registry holds every trained model, one entry per city.



CI/CD AUTOMATION

GitHub Actions

Feature collection every hour, model retraining every day authenticated keylessly via Workload Identity Federation.



DASHBOARD

Streamlit

An interactive dashboard with city selection, SHAP explainability, and hazardous-AQI alerts, deployed on Cloud Run.



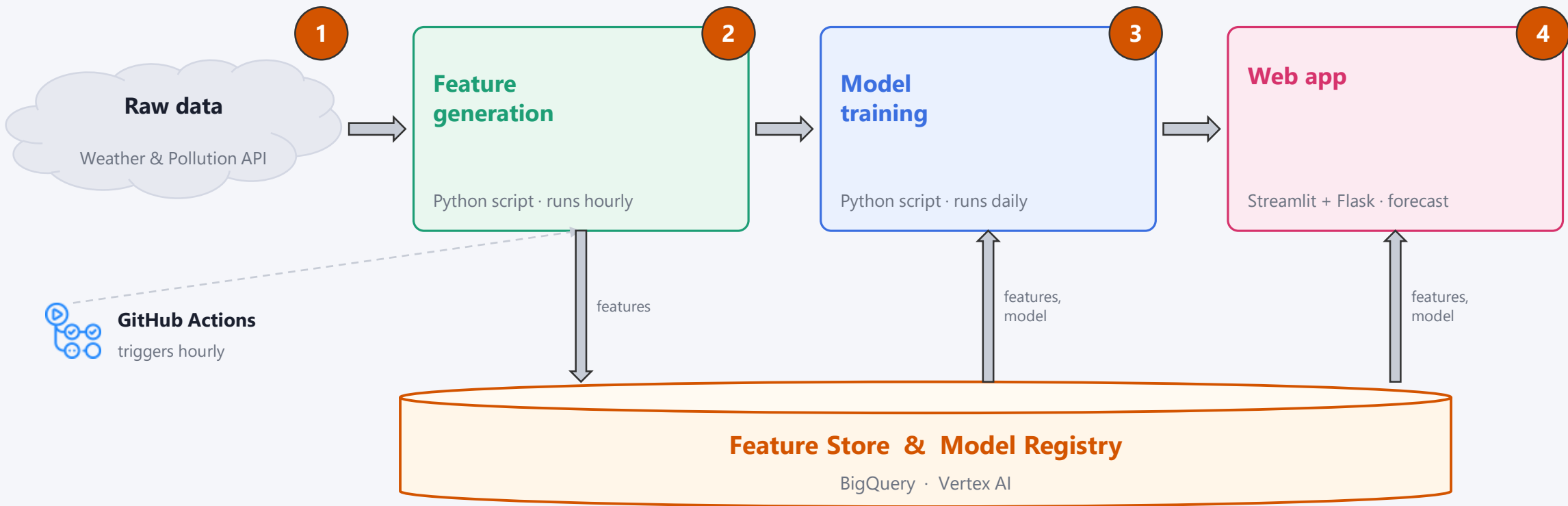
REST API

Flask

A JSON API exposing the same forecasts programmatically, deployed as its own serverless Cloud Run service.

End-to-End Architecture

Air Quality Index (AQI) prediction service — as defined in the project brief



Six cities · 100% serverless — every component scales to zero and costs nothing when idle

Results & Live Demo

Everything below is live right now, running unattended

Live links

Dashboard

aqi-predictor-dashboard-444682761540.asia-south1.run.app

GitHub repo

https://github.com/Naveedjokhio/10Pearls_aqi-predictor

Also implemented: SHAP explainability, hazardous-AQI alerts, city ranking, forecast-confidence indicators, and a raw-data export.

Sample results of Karachi

Horizon	Best model	RMSE	R ²
Day 1	Neural Net	6.9	0.75
Day 2	Random Forest	7.8	0.66
Day 3	Neural Net	7.0	0.72

All three model families (Ridge, Random Forest, TensorFlow) are trained per city and per horizon; the median of their predictions is used for the final forecast, guarding against any single model's outliers.